



Name: _____ Cohort/Batch: _____ Date: _____ Score: _____

RASPBERRY PI PRACTICE SHEET - 10 CONCEPTUAL Q&A

10 conceptual Q&A IoT & Edge

TOTAL: 10 QUESTIONS

Q1 What is Raspberry Pi and what problem in IoT & Edge does it solve?

Q6 How does Raspberry Pi handle reliability and high availability?

Q2 What are the core components or concepts in Raspberry Pi?

Q7 What observability does Raspberry Pi provide out of the box?

Q3 How does Raspberry Pi compare to its main alternatives?

Q8 What are common performance bottlenecks in Raspberry Pi?

Q4 How do you get started with Raspberry Pi for a new project?

Q9 What security best practices apply when running Raspberry Pi?

Q5 What configuration options most affect Raspberry Pi's behavior in production?

Q10 What mistakes do teams commonly make when adopting Raspberry Pi?



ANSWER KEY & EXPLANATORY SOLUTIONS

Comprehensive verified answers, best-practice configs, security controls & reference

VERIFIED SOLUTIONS

A1 Raspberry Pi is a tool or platform

Mitigation

Raspberry Pi is a tool or platform that automates or simplifies a specific concern in the IoT & Edge domain, reducing manual effort and operational complexity for engineering teams.

A6 Raspberry Pi provides HA through leader election,

Observability

Raspberry Pi provides HA through leader election, replication, or stateless horizontal scaling. Deploy at least two instances across availability zones and configure health checks for automatic failover.

A2 Raspberry Pi has key building blocks that

Primitives

Raspberry Pi has key building blocks that work together: a configuration layer defines intent, a runtime layer enforces it, and an API or CLI layer allows operators to manage and observe the system.

A7 Raspberry Pi exposes metrics (Prometheus endpoint or

Automation

Raspberry Pi exposes metrics (Prometheus endpoint or built-in dashboard), structured logs, and optionally distributed traces. Define SLOs on the key metrics and alert before users are affected.

A3 Raspberry Pi trades off simplicity against feature

Hardening

Raspberry Pi trades off simplicity against feature richness differently than its alternatives. Choose Raspberry Pi when its specific strengths performance, ecosystem, or ease of use align with your team's primary constraints.

A8 Performance bottlenecks typically stem from misconfigured

Performance

Performance bottlenecks typically stem from misconfigured connection pools, large payload sizes, insufficient worker threads, or missing caches. Profile with built-in diagnostics before optimizing.

A4 Install Raspberry Pi via its package manager

Gotchas

Install Raspberry Pi via its package manager or binary release. Follow the quickstart to initialize a project, configure the minimal required settings, and run the default command to verify the setup works end-to-end.

A9 Run Raspberry Pi with the principle of

Assessment

Run Raspberry Pi with the principle of least privilege, enable TLS for all communication, use a secrets manager for credentials, and keep the software patched to the latest stable release.

A5 Production-critical settings typically include concurrency

Concurrency

Production-critical settings typically include concurrency limits, timeout values, retry policies, resource limits, and logging levels. Review the official production hardening guide before deploying.

A10 Common pitfalls include skipping the documentation and

Documentation

Common pitfalls include skipping the documentation and learning the API by trial and error, using default configurations in production, lacking a runbook for operational issues, and not testing failover scenarios.